Section D
Group 8

US Real Estate Analysis

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Sector
Real Estate

Context & Problem Statement

REAL ESTATE MARKET CONTEXT

Real estate is one of the largest asset classes, where buyers, investors, and analysts regularly evaluate properties based on pricing, location, and features to make investment and purchasing decisions.

PROBLEM STATEMENT

How can buyers and investors make reliable property decisions and identify fair pricing in a competitive real estate market?

OBJECTIVE

Develop a structured and reliable system that enables accurate property comparison and supports better pricing and investment decisions.

Data Engineering & Pipeline

FROM RAW DATA TO ANALYTICS-READY DATASET

01 DATA SOURCE

- US Real Estate Listings (Kaggle dataset)
- ~600,000 rows × 28 column
- features Covers multiple US states and property types
- Includes pricing, location, and property details

02 DATA PREPARATION

- Removed invalid records (e.g., incorrect values)
- Standardized units (acres → sqft)
- Selected relevant features for analysis
- Prepared consistent dataset for processing

03 DATA CLEANING

- Handling missing values (hierarchical imputation)
- Fixing inconsistencies in key attributes
- Outlier treatment (price & area capping)
- Ensured logical consistency across features

04 FINAL DATASET

- 59,581 records with 19 features
- No missing or invalid values
- Engineered features (Price per SqFt)
- Ready for analysis, modeling, and dashboards

DATA SOURCE: https://www.kaggle.com/datasets/polartech/500000-us-homes-data-for-sale-properties

KPI & Metrics Framework

KEY PERFORMANCE INDICATORS FOR REAL ESTATE DATA ANALYSIS

01

MEDIAN SALE PRICE

$ 439,990

Typical market price
Used to assess affordability

= Median(Sale Price)

02

PRICE PER SQFT

$ 221

Standardized pricing metric
Enables fair comparison

=Sale Price ÷ Living Area

03

STATUS RATIO

55.9% For Sale

Shows market demand
Indicates listing activity

= (For Sale Listings ÷ Total Listings) × 100

04

TOTAL LISTINGS

59,581

Market size indicator
Represents dataset coverage

= Count of Unique Properties

05

PROPERTY TYPE MIX

80% Single Family

Shows market composition
Helps segment analysis

=(Count of Property Type ÷ Total Listings) × 100

Key Insights from EDA

Single-Family Homes Dominate the Market

Insight: Majority of listings are single-family homes, forming the core of market supply
Impact: Best segment for stable investment and consistent pricing benchmarks

Property Value Varies Strongly by Location

Insight: Significant price differences exist across states and cities
Impact: Location is a primary driver of valuation and investment strategy

Outliers Distort Market Perception

Insight: A small number of luxury properties inflate average prices
Impact: Outlier handling is critical for realistic pricing insights

Market Activity is Supply-Heavy

Insight: Majority of listings are actively for sale
Impact: Indicates competitive market conditions for buyers and investors

Detailed analysis and visualizations available in EDA Notebook (03_eda.ipynb)

Advanced Analysis

STRONG PRICE VARIATION EXPLAINED

MARKET SEGMENTATION
Identified 3 distinct property tiers: Budget, Mid-Range, and Luxury

PRICE DRIVERS
Latitude/Longitude and Living Space emerge as top price predictors.

MODEL INSIGHT
Statistical modeling captures key factors influencing price variation

01 **STATISTICAL VALIDATION**

Validated price differences across property types using hypothesis testing (significant at $p < 0.05$)

02 **PREDICTIVE MODELING**

Regression analysis quantified how key features like space and location influence pricing

03 **MARKET SEGMENTATION**

K-Means clustering grouped properties into distinct investment segments

04 **TOOLS & METHODS**

Used Scikit-learn, Statsmodels, and SciPy for modeling and statistical testing, supported by Matplotlib for visualization

EXECUTIVE VIEW (GEOGRAPHIC)

Interactive map highlighting price hotspots across states, enabling quick identification of high-value and emerging markets.

OPERATIONAL VIEW (PROPERTY)

Detailed breakdown of property types, bedroom distribution, and listing status, supporting inventory and supply-demand analysis.

DEEP DIVE (PRICING ANALYSIS)

Advanced analysis of price drivers including living space, land size, and $/SqFt, uncovering patterns across regions.

STAKEHOLDER INTERACTIVITY

Dynamic filters and drill-down capabilities allow users to explore data by state, city, and property type in real time.

Dashboard Walkthrough

Designed to support both strategic decision-making and detailed analysis, enabling users to move from high-level insights to granular exploration.

https://public.tableau.com/app/profile/anant.jain2510/viz/DVA_capstone_tableau/Dashboard1

Recommendations

01

Surgical Entry

Insight: Mid-range segment shows balanced pricing and consistent demand
Recommendation: Focus investments on mid-range properties in high-activity markets
Impact: Better return consistency with controlled risk exposure

02

Targeting Efficiency

Insight: Single-family homes dominate listings and show stable pricing patterns
Recommendation: Prioritize single-family properties for core portfolio allocation
Impact: Improved liquidity and more reliable valuation benchmarks

03

Market Access Strategy

Insight: A small number of agencies control a large share of listings
Recommendation: Build partnerships with high-volume real estate agencies
Impact: Faster access to quality listings and improved deal pipeline

04

Pricing Discipline

Insight: Price per SqFt provides a consistent benchmark across markets
Recommendation: Use PPSF benchmarks to evaluate and compare property value
Impact: Better pricing decisions and reduced risk of overpaying

Impact & Value

Operational Impact

- Reduced manual data preparation effort through automated ETL pipeline
- Enabled faster turnaround from raw data to analysis-ready dataset

Data Quality Improvement

- Standardized key features like pricing, location, and property attributes
- Improved consistency, enabling fair comparison across listings

Decision-Making Value

- Introduced reliable benchmarks such as Price per SqFt for valuation
- Enabled stakeholders to make more confident and consistent investment decisions

Business Relevance (So What?)

- Supports better property selection and pricing evaluation
- Reduces risk of mispricing and inefficient capital allocation
- Provides a scalable foundation for future analytics and modeling

Limitations & Next Steps

Current Scope

- Analysis is based on listing-level data up to 2022
- Focused on property attributes and pricing features
- Limited integration of external market indicators

Key Constraints

- Absence of transaction-level data limits price validation
- Market dynamics (interest rates, demand shifts) are not fully captured
- Insights reflect historical patterns rather than real-time conditions

Next Steps

- Develop predictive pricing models using machine learning
- Build real-time dashboards for continuous market monitoring
- Integrate external datasets (interest rates, demographics, rental trends)
- Extend pipeline for live data ingestion and automated updates

Thank You!

We appreciate your attention and interest!

Section D — Group 8

github.com/section-d-group8

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Questions & Discussion
We welcome your questions and look forward to discussing insights, methodology, and potential applications of this work.